

Music Player System Project Report

1. Preface

This report presents the development and implementation of the Music Player System project developed during the Core Java Internship program.

2. Introduction

The Music Player System is a Core Java based application developed for managing songs and playlists.

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd is a technology-focused organization providing internship opportunities and practical learning experiences for students.

2.2 About upskill Campus

upskill Campus is an online learning and internship platform that helps students gain industry-level practical skills.

2.3 Objective

The objective of this project is to develop a Music Player application using Core Java concepts such as classes, objects, methods, and collections.

2.4 Reference

Oracle Java Documentation, Java Tutorials, Core Java Books, and online learning resources were referred during project development.

2.5 Glossary

Core Java: Fundamental Java concepts.

ArrayList: Dynamic collection framework in Java.

OOP: Object-Oriented Programming.

3. Problem Statement

Develop a Music Player application that allows users to manage songs and playlists with playback functionality.

4. Existing and Proposed Solution

Existing systems are complex and feature-rich. The proposed solution is a simple and user-friendly Core Java application for music playback and playlist management.

5. Proposed Design/Model

The project follows a modular structure using separate classes for songs, playlists, and music player operations.

5.1 High Level Diagram

The application consists of modules such as Main Class, Music Player Module, Song Module, and Playlist Module.

5.2 Low Level Diagram

Low-level design includes class interactions between Main.java, Song.java, PlaylistManager.java, and MusicPlayer.java.

5.3 Interfaces

The application currently uses console-based interaction for user operations.

6. Performance Test

The project was tested for playlist creation, playback operations, and song management functionality.

6.1 Test Plan/Test Cases

Test cases included adding songs, removing songs, displaying playlists, and playback operations.

6.2 Test Procedure

The application was executed using Java JDK and tested through console outputs.

6.3 Performance Outcome

The application executed successfully and produced correct playlist and playback outputs.

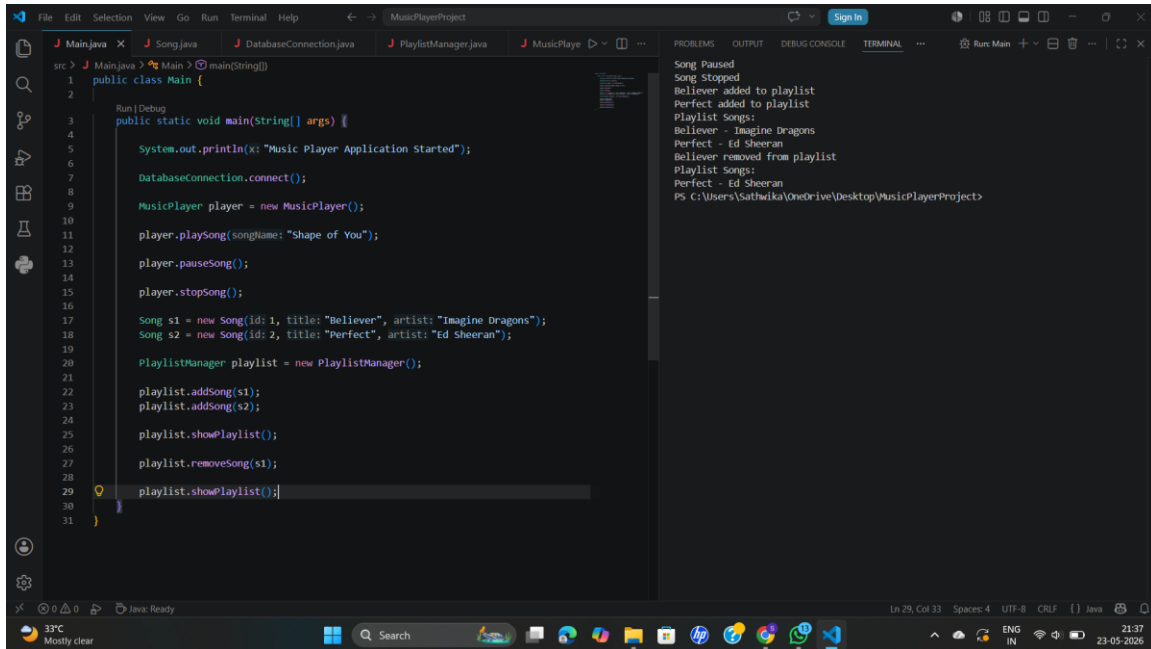
7. My Learnings

This project improved my understanding of Core Java concepts, collections, classes, objects, methods, and project structuring.

8. Future Work Scope

Future enhancements may include GUI development using Swing, audio support, JDBC database connectivity, and advanced music search functionality.

Output Screenshots



The screenshot displays an IDE window for a project named "MusicPlayerProject". The editor shows the following Java code in `Main.java`:

```
1 public class Main {
2
3     public static void main(String[] args) {
4
5         System.out.println("Music Player Application Started");
6
7         DatabaseConnection.connect();
8
9         MusicPlayer player = new MusicPlayer();
10
11         player.playSong(songName: "Shape of You");
12
13         player.pauseSong();
14
15         player.stopSong();
16
17         Song s1 = new Song(id: 1, title: "Believer", artist: "Imagine Dragons");
18         Song s2 = new Song(id: 2, title: "Perfect", artist: "Ed Sheeran");
19
20         PlaylistManager playlist = new PlaylistManager();
21
22         playlist.addSong(s1);
23         playlist.addSong(s2);
24
25         playlist.showPlaylist();
26
27         playlist.removeSong(s1);
28
29         playlist.showPlaylist();
30
31     }
```

The right-hand pane shows the output of the program:

```
Song Paused
Song Stopped
Believer added to playlist
Perfect added to playlist
Playlist Songs:
Believer - Imagine Dragons
Perfect - Ed Sheeran
Believer removed from playlist
Playlist Songs:
Perfect - Ed Sheeran
PS C:\Users\sathwika\OneDrive\Desktop\MusicPlayerProject>
```

The status bar at the bottom indicates the file is at Line 29, Column 33, using UTF-8 encoding with CRLF line endings. The system tray shows a temperature of 33°C and the date 23-05-2026.